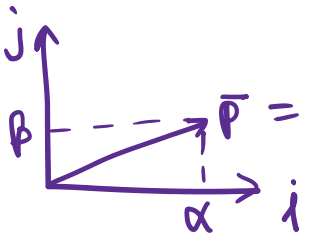
DTFS:

$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n} \longrightarrow \text{F.S. Repres}^n \text{ of } x[n]$$



$\bar{p} = \alpha i + \beta j$, $\alpha = \langle p, i \rangle$
$\beta = \langle p, j \rangle$

$\{x_k\}_{k=0}^{N-1}$ is o.n. basis of S_N , $x_k[n] \triangleq e^{jk\frac{2\pi}{N}n}$

$\Rightarrow x = \sum a_k x_k$, Let $p \in [0, N-1]$,

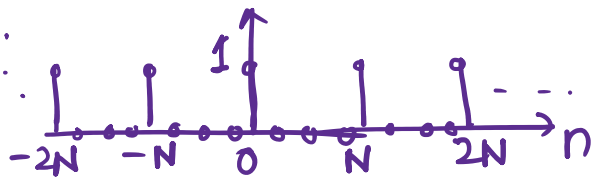
$$\langle x, x_p \rangle = \langle \sum_k a_k x_k, x_p \rangle = \sum_k a_k \langle x_k, x_p \rangle = a_p$$

$$\therefore a_k = \langle x, x_k \rangle = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\frac{2\pi}{N}n} \longrightarrow \text{F.S. Coeff.} \Big\} \text{Analysis}$$

$$\& \; x[n] = \sum_{k=0}^{N-1} a_k \underbrace{e^{jk\frac{2\pi}{N}n}}_{\text{diff. freq. components}} \longrightarrow \text{F.S. Repres}^n \Big\} \text{Synthesis}$$

strength of diff. freq. components

Ex: $x[n]$:



$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n}$$

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\frac{2\pi}{N}n}$$

$$= 1/N \; , \forall k$$

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① $a_{k+N} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j(k+N)\frac{2\pi}{N}n} = a_k$

② For real $x[n]$, $a_{-k} = a_k^*$

Pf: $x^*[n] = x[n]$

$$\sum_{k'=(N-1)}^{0} a_{k'}^* e^{jk'\frac{2\pi}{N}n} = \sum_{k=0}^{N-1} a_{-k}^* e^{jk\frac{2\pi}{N}n}$$

$$\sum_{k=0}^{N-1} a_k^* e^{-jk\frac{2\pi}{N}n} = \sum_{k'=-(N-1)} a_{k'}^* e^{jk'\frac{2\pi}{N}n} = \sum_{k=0} a_{-k}^* e^{jk\frac{2\pi}{N}n}$$

Let $-k = k'$

$$= \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n} \quad (= x[n])$$

$$\therefore a_k = a_{-k}^* \Rightarrow a_{-k} = a_k^*$$

③ $$\sum_{k=0}^{N-1} |a_k|^2 = \frac{1}{N}\sum_{n=0}^{N-1} |x[n]|^2$$

Pf: $$\sum_{k=0}^{N-1} a_k a_k^* = \sum_{k=0}^{N-1} a_k \cdot \frac{1}{N}\sum_{n=0}^{N-1} x^*[n] e^{j\frac{2\pi}{N}nk}$$

$$= \frac{1}{N}\sum_{n=0}^{N-1} x^*[n] \underbrace{\sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n}}$$

$$= \frac{1}{N}\sum_{n=0}^{N-1} x^*[n]\, x[n] = \frac{1}{N}\sum_{n=0}^{N-1} |x[n]|^2$$

Continuous Time Fourier Series (CTFS)

$$x(t) = \sum_{k=} a_k e^{jk\frac{2\pi}{T}t}$$

$$S_T = \left\{ x : \mathbb{R} \to \mathbb{C} \;\middle|\; x(t+T) = x(t)\ \forall t \ \&\ \frac{1}{T}\int_{\langle T \rangle} |x(t)|^2 dt < \infty \right\}$$